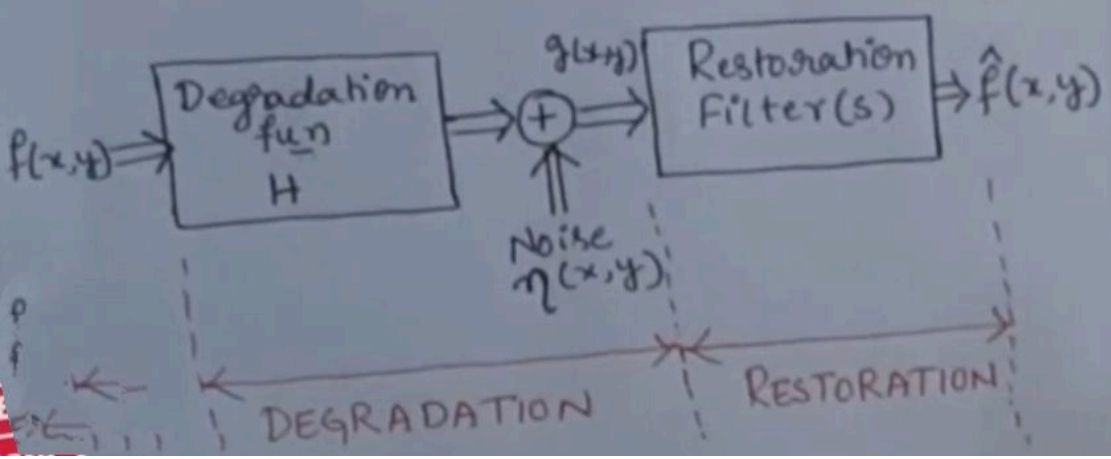
Image Restoration:

→ Reconstruct (or) Recover an Image
→ degraded → using Prior Knowledge

→ Restoration → modeling the degradation
→ Applying Inverse process → recover

→ Restoration techniques.
(i) Spatial domain (ii) Frequency domain.
· Additive Noise Image blur.

A Model of Image Degradation | Restoration Process:



$$g(x,y) = h(x,y) * f(x,y) + \eta(x,y) \rightarrow (1)$$

$$G(u,v) = H(u,v)\,F(u,v) + N(u,v) \rightarrow (2)$$

The Principle Use of Noise Models

- **Gaussian noise:**
  - electronic circuit noise and sensor noise due to poor illumination or high temperature.
- **Rayleigh noise:**
  - Noise in range imaging.
- **Erlang noise:**
  - Noise in laser imaging.
- **Impulse noise:**
  - Quick transients take place during imaging.
- **Uniform noise:**
  - Used in simulations.

Important Noise Probability Density fun: [PDF]

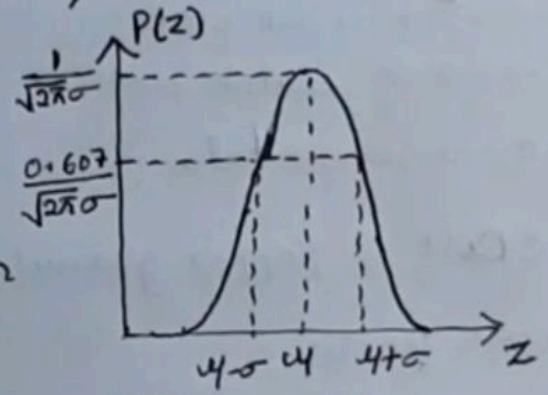
Noise → Image acquisition and/or transmission.

→ Spatial properties of Noise.

→ Frequency properties of Noise.

(i) Gaussian Noise: [Normal noise model]

PDF, $P(z) = \frac{1}{\sqrt{2\pi}\,\sigma} e^{-(z-\mu)^2/2\sigma^2}$



z → graylevel

μ → mean of avg of z

σ → Standard deviation

σ^2 → Variance.

70% of values → $[(\mu-\sigma), (\mu+\sigma)]$

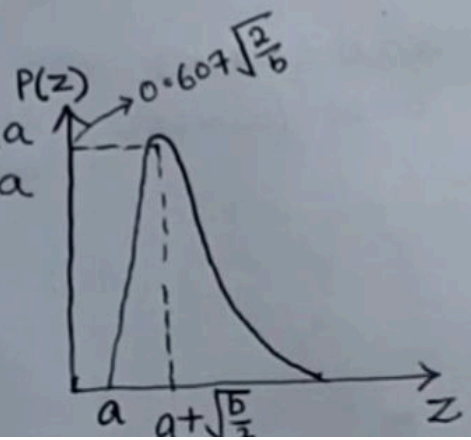
95% of values → $[(\mu-2\sigma), (\mu+2\sigma)]$

(ii) Rayleigh noise:

$$P(z) = \begin{cases} \frac{2}{b}(z-a)e^{-(z-a)^2/b} & ; z \geq a \\ 0 & ; z < a \end{cases}$$

mean: $\mu = a + \sqrt{\pi b/4}$

Variance: $\sigma^2 = \frac{b(4+\pi)}{4}$

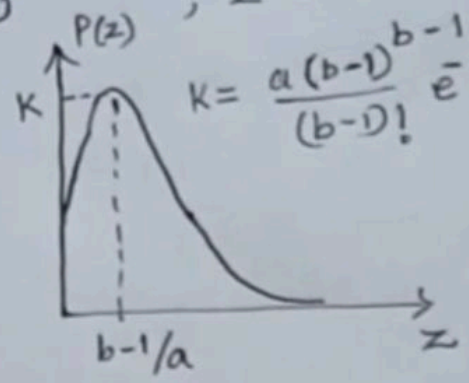


(iii) Erlang [Gamma] noise:

$$P(z) = \begin{cases} \frac{a^b z^{b-1}}{(b-1)!} e^{-az} & ; z \geq 0 \\ 0 & ; z < 0 \end{cases}$$

mean: $\mu = \frac{b}{a}$

Variance: $\sigma^2 = \frac{b}{a^2}$



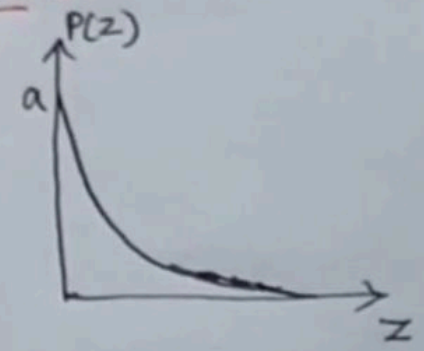
$K = \frac{a(b-1)^{b-1}}{(b-1)!} e^{-(b-1)}$

(iv) Exponential noise:

$$P(z) = \begin{cases} a \cdot e^{-az} & ; z \geq 0 \\ 0 & ; z < 0 \end{cases}$$

$\mu = \frac{1}{a}$

$\sigma^2 = \frac{1}{a^2}$

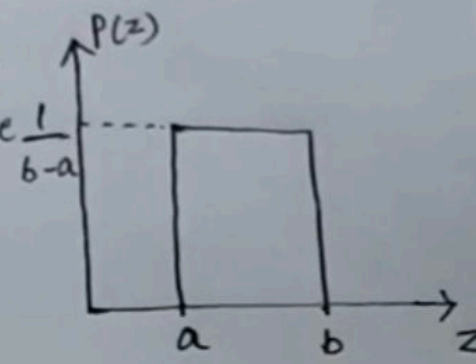


(v) Uniform noise:

$$P(z) = \begin{cases} \frac{1}{b-a} & ; a \leq z \leq b \\ 0 & ; \text{otherwise} \end{cases}$$

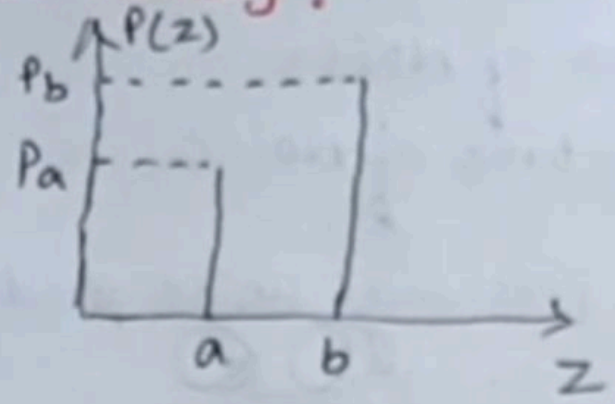
$\mu = \frac{a+b}{2}$

$\sigma^2 = \frac{(b+a)^2}{12}$



(vi) Salt & Pepper noise [Impulse noise]:

$$P(z) = \begin{cases} P_a & ; z = a \\ P_b & ; z = b \\ 0 & ; \text{otherwise} \end{cases}$$



→ $b > a \Rightarrow$ gray level b → light dot
gray level a → dark dot

→ $P_a = 0$ (or) $P_b = 0$ → unipolar

→ $P_a \cong P_b$ → Salt & pepper granules.

Shot & Spike noise.

−ve impulses → black (Pepper) point
+ve impulses → white (salt) point.

8-bit ⇒ $a = 0$ (black)
$b = 255$ (white)

Restoration in Presence of Noise Only –

Spatial Filtering:

Degradation → Only due to noise.

$$g(x,y) = f(x,y) + \eta(x,y)$$

$$\& \; G(u,v) = F(u,v) + N(u,v)$$

I. MEAN FILTERS:

(i) Arithmetic mean Filter:

→ Simplest mean Filter.

$$\boxed{\hat{f}(x,y) = \frac{1}{mn} \sum_{(s,t) \in S_{xy}} g(s,t)} \longrightarrow (i)$$

S_{xy} → set of co-ordinates in a rectangular subimage window of size $m \times n$. Centered at point (x,y)

→ Avg of corrupted image $g(x,y)$ in the area defined by S_{xy}.

→ Smoothens local variation

→ Noise is reduced due to blurring.

(ii) Geometric mean Filter:

$$\boxed{\hat{f}(x,y) = \left[\prod_{(s,t) \in S_{xy}} g(s,t) \right]^{\frac{1}{mn}}} \longrightarrow (ii)$$

→ Product of pixels in the subimage window → raised to power $\frac{1}{mn}$

→ Smoothing simillar to arithmetic mean filter.

→ lose less image details.

(iii) Harmonic mean Filter:

$$\boxed{\hat{f}(x,y) = \frac{mn}{\sum_{(s,t) \in S_{xy}} \frac{1}{g(s,t)}}} \longrightarrow (iii)$$

→ Works well for Salt noise & Fails for Pepper noise

→ also works well for Gaussian noise.

(iv) Contra harmonic mean Filter:

$$\boxed{\hat{f}(x,y) = \frac{\sum_{(s,t)\in S_{xy}} g(s,t)^{Q+1}}{\sum_{(s,t)\in S_{xy}} g(s,t)^{Q}}}$$

$Q \rightarrow$ order of the filter.

→ Reducing the effect of Salt & pepper noise

→ $Q \rightarrow +ve \rightarrow$ it eliminates pepper noise

$Q \rightarrow -ve \rightarrow$ it eliminates salt noise.

→ Can not do both simultaneously.

→ Reduces to Arithmetic mean filter if $Q = -1$.

Restoration in presence of noise only - Spatial Filtering:

II. ORDER-STATISTICS FILTERS:

→ Response → ordering (ranking) the pixels.
→ determined by the ranking results.

(i) Median Filter:

$$\boxed{\hat{f}(x,y) = \underset{(s,t)\in S_{xy}}{\text{median}}\{g(s,t)\}} \longrightarrow (i)$$

→ Replaces the value of pixel by the median of gray levels

→ Excellent noise reduction with less blurring

→ effective in presence → bipolar & Unipolar Impulse noise.

(ii) Min & Max Filter:

$$\boxed{\hat{f}(x,y) = \max_{(s,t)\in S_{xy}}\{g(s,t)\}} \longrightarrow (ii)$$

Pepper noise ↓

$$\boxed{\hat{f}(x,y) = \min_{(s,t)\in S_{xy}}\{g(s,t)\}} \longrightarrow (iii)$$

Salt noise ↓

max → Finding the brightest point
min → Find the darkest point.

(iii) Mid Point Filter:

$$\boxed{\hat{f}(x,y) = \frac{1}{2}\left[\max_{(s,t)\in S_{xy}}\{g(s,t)\} + \min_{(s,t)\in S_{xy}}\{g(s,t)\}\right]} \longrightarrow (iv)$$

→ Computes the mid point b/w max & min values.

→ Combines order statistic & averaging

→ works best → Randomly distributed noise

(iv) Alpha-trimmed mean Filter:

$$\boxed{\hat{f}(x,y) = \frac{1}{mn-d}\sum_{(s,t)\in S_{xy}} g_r(s,t)}$$

$d \to 0$ to mn

When $d=0$ → Reduced to Arith mean Filt.

$d = \frac{(mn-1)}{2}$ → median Filter.

Other values → useful in multiple type of noise reduction

Periodic Noise Reduction using Freq domain Filtering:

(i) Band Reject Filters:

→ Attenuate a band of frequencies above the origin of the F.T.

Ideal

$$H(u,v) = \begin{cases} 1 \; ; \; D(u,v) < D_0 - \frac{W}{2} \\ 0 \; ; \; D_0 - \frac{W}{2} \le D(u,v) \le D_0 + \frac{W}{2} \\ 1 \; ; \; D(u,v) > D_0 + \frac{W}{2} \end{cases} \rightarrow (i)$$

$D(u,v)$ → distance from the origin of the Centered freq band.

W → width of the band

D_0 → radial center. / cutoff freq

III^ly Butterworth

$$H(u,v) = \frac{1}{1 + \left[\frac{D(u,v)W}{D^2(u,v) - D_0^2}\right]^{2n}} \rightarrow (ii)$$

n → order

& Gaussian

$$H(u,v) = 1 - e^{-\frac{1}{2}\left[\frac{D^2(u,v) - D_0^2}{D(u,v)W}\right]} \rightarrow (iii)$$

(ii) Bandpass Filters:

→ opposite operation of a bandreject

$$H_{bp}(u,v) = 1 - H_{br}(u,v) \rightarrow (i)$$

(iii) Notch Filters:

→ pauses Frquencies in predefined neighborhoods.

Ideal notch Reject.

radius D_0 & centres at (u_0, v_0) with Symmetry at $(-u_0, -v_0)$ is

$$H(u,v) = \begin{cases} 0 \; ; \; D_1(u,v) \le D_0 \text{ (or) } D_2(u,v) \le D_0 \\ 1 \; ; \; \text{otherwise} \end{cases} \rightarrow (i)$$

where; $D_1(u,v) = \left[\left(u - M/2 - u_0\right)^2 + \left(v - N/2 - v_0\right)^2\right]^{1/2}$

& $D_2(u,v) = \left[\left(u - M/2 + u_0\right)^2 + \left(v - N/2 + v_0\right)^2\right]^{1/2}$

Butter worth 'n'

$$H(u,v) = \frac{1}{1 + \left[\frac{D_0^2}{D_1(u,v) D_2(u,v)}\right]^n} \rightarrow (ii)$$

Gaussian $H(u,v) = 1 - e^{-\frac{1}{2}\left[\frac{D_1(u,v) D_2(u,v)}{D_0^2}\right]} \rightarrow (iii)$

High pass Filters ⇒ $u_0 = v_0 = 0$

$$H_{hp}(u,v) = 1 - H_{hr}(u,v) \rightarrow (iv)$$

Degradation models:

- Many types of degradation can be approximated by linear, space invariant processes
  - Can take advantages of the mature techniques developed for linear systems
- Non-linear and space variant models are more accurate
  - Difficult to solve
  - Unsolvable

Estimating degradation function

- Estimation by image observation
  - Degradation system H is completely characterized by its impulse response
  - Select a small section from the degraded image $g_S(x, y)$
  - Reconstruct an unblurred image of the same size $\hat{f}_S(x, y)$
  - The degradation function can be estimated by

$$H_s(u,v) = \frac{G_s(u,v)}{\hat{F}_s(u,v)}$$

By ignoring the noise term, G(u,v) = F(u,v)H(u,v). If F(u,v) is the Fourier transform of point source (impulse), then G(u,v) is approximates H(u,v).

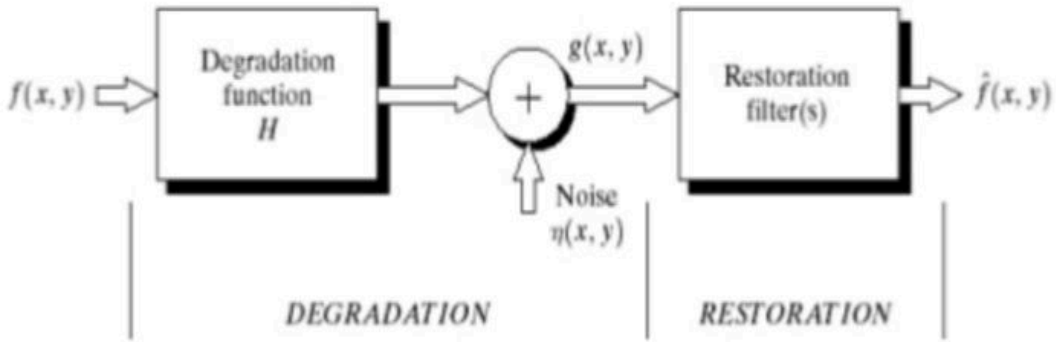


Fig: A model of the image degradation / restoration process

Continuous degradation model

- Motion blur. It occurs when there is relative motion between the object and the camera during exposure.

$$h(i) = \begin{cases} \frac{1}{L}, & \text{if } -\frac{L}{2} \le i \le \frac{L}{2} \\ 0, & \text{otherwise} \end{cases}$$

- Atmospheric turbulence. It is due to random variations in the reflective index of the medium between the object and the imaging system and it occurs in the imaging of astronomical objects.

$$h(i, j) = K \exp\left(-\frac{i^2 + j^2}{2\sigma^2}\right)$$

- Uniform out of focus blur

$$h(i, j) = \begin{cases} \frac{1}{\pi R}, & \text{if } \sqrt{i^2 + j^2} \le R \\ 0, & \text{otherwise} \end{cases}$$

- Uniform 2-D blur

$$h(i, j) = \begin{cases} \frac{1}{(L)^2}, & \text{if } -\frac{L}{2} \le i, j \le \frac{L}{2} \\ 0, & \text{otherwise} \end{cases}$$

Inverse filtering:

- Degradation model

$$g(x,y) = f(x,y) \otimes h(x,y) + \eta(x,y)$$
$$G(u,v) = F(u,v)H(u,v) + N(u,v)$$

This expression tells us that even if we know the degradation function we cannot recover the undegraded image exactly because of the random noise, whose Fourier transform is not known. Image restoration is an ill-posed problem. When H (u, v) is very small, the noise term dominates the restoration result.

- Inverse filter

$$\hat{F}(u,v) = \frac{G(u,v)}{H(u,v)} + \frac{N(u,v)}{H(u,v)}$$
$$= F(u,v) + \frac{N(u,v)}{H(u,v)}$$

Assume h is known (low-pass filter)

Inverse filter G(u,v) = 1 / H(u,v)

$$\tilde{X}(u,v) = Y(u,v)\, G(u,v)$$

Problems with Inverse Filtering

H(u,v) = 0, for some u, v

In noisy case, $y = x * h + n$

n: additive noise

The simplest approach to restoration is direct inverse filtering, where we compute an estimate, f(u,v), of the transform of the original image simply by dividing the transform of the degraded image, G(u,v), by the degradation function:

$$F(u,v) = \frac{G(u,v)}{H(u,v)}$$

$$F(u,v) = F(u,v) + \frac{N(u,v)}{H(u,v)}$$

This is an interesting expression. It tells us that even if we know the degradation function we cannot recover the undegraded image exactly because N(u,v) is a random function whose Fourier transform is not known.

There is more bad. News. If the degradation has zero or very small values, then the ratio N(u,v)/H(u,v) could easily dominate the estimate F(u,v). this in fact is frequently the case.

One approach to get around the zero or small value problem is to limit the filter frequencies to values near the origin.

$$F(0,0) = \frac{1}{MN}\sum_{x=0}^{M-1}\sum_{y=0}^{N-1} f(x,y)$$

By this equation we known that H(0,0) is equal to the average value of h(x,y) and that this is usually the highest value of H(u,v) in the frequency domain. Thus by limiting the analysis to frequencies near the origin, we reduce the probability of encountering zero values.